

Classical Mechanics

$$W = \vec{F} \cdot d\vec{r}$$

$$(\vec{F}_c) d\vec{r}$$

Force of constraint is missing.

$$(\vec{F}_e + \vec{F}_c) \cdot d\vec{r}$$

$$\frac{d^2 \vec{r}}{dt^2} = (\vec{F}_e + \vec{F}_c)$$

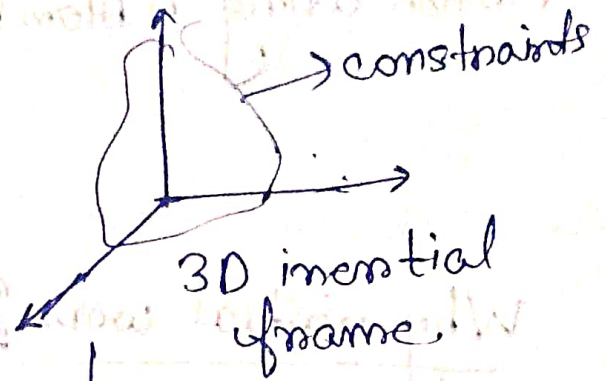
Virtual work

defined as

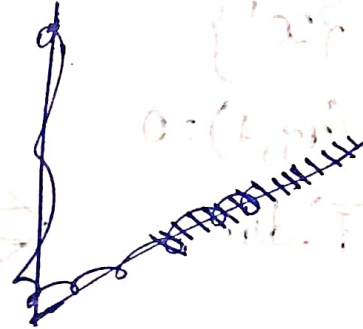
virtual displacement from one pt. to another
(displacement of a system keeping everything frozen (time) also)

transformation
constraints

generalized coordinates
(constraints will not appear here).



configuration space,



Assignment

Derive the ~~equations of~~ Lagrange of equation of Motion using D'Alembert's Principle.

→ Submit - Friday.

20/08/2025

Why virtual work?

Inertial Frame

$$\begin{aligned} &\vec{r}_i \\ &\{x^i\} \\ &f(\vec{r}_i, t) = 0 \\ &\vec{F} \cdot d\vec{r} \end{aligned}$$

Generalized Coordinates

$$\begin{aligned} &\{q_i\} \\ &\text{no constraint} \end{aligned}$$

$$\begin{aligned} &Q_i dq_i \\ &\hookrightarrow Q_i \text{ is generalized force.} \end{aligned}$$

$$\vec{F} = \sum_i (F_i \hat{x}_i) = \sum_i Q_i \hat{q}_i$$

$$\begin{aligned} x^i &= x^i(q_j, t) \\ dx^i &= \sum_j \frac{\partial x^i}{\partial q_j} dq_j + \frac{\partial x^i}{\partial t} dt \end{aligned}$$

$$dW = \sum_{i,j} F_i \frac{\partial x^i}{\partial q_j} dq_j + \sum_i F_i \frac{\partial x^i}{\partial t} dt = \sum_j Q_j dq_j +$$

$$\delta x^i = \frac{\partial x^i}{\partial q_j} \delta q_j + \left(\dots \right)$$

→ Time is not frozen → instantaneously calculate in δt .

• A particle of mass 'm' is confined to a circle which is expanding with time and subjected to a force $\vec{F} = F_x \hat{x} + F_y \hat{y}$.

$$x^2 + y^2 = R^2(t).$$

$$\Rightarrow x = R(t) \cos \theta.$$

$$y = R(t) \sin \theta.$$

$$dx = -R(t) \sin \theta d\theta + \frac{\partial R}{\partial t} \cos \theta dt.$$

$$dy = R(t) \cos \theta d\theta + \frac{\partial R}{\partial t} \sin \theta dt.$$

$$dW = F_i dx^i = F_x \left[-R(t) \sin \theta d\theta + \frac{\partial R}{\partial t} \cos \theta dt \right] + F_y \left[R \cos \theta d\theta + \frac{\partial R}{\partial t} \sin \theta dt \right].$$

$$dW = \underbrace{[-F_x \sin \theta + F_y \cos \theta]}_{\text{form of } Q} R d\theta + \underbrace{[F_x \cos \theta + F_y \sin \theta]}_{\Pi} \frac{\partial R}{\partial t} dt$$

Work should be in the form of $Q d\theta$. But here additional Π term, which we cannot equate to 0 as $dR \neq 0$ and F_x and $F_y \neq 0$ as they are function of x and y and $\theta \neq 0$ $\sin \theta, \cos \theta \neq 0$ always. → This is a contradiction.

Lagrange Equation of Motion

① We have generalized coordinate $(q_1, q_2, \dots, q_n)$.

② Kinetic energy $T(q_i, \dot{q}_i, t) \rightarrow \frac{1}{2} m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2)$.

③ Potential $V(q_1, q_2, \dots, t)$, $F_j = -\frac{\partial V}{\partial q_j}$.

Define

$$L = T - V.$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} = 0.$$

$\rightarrow$ 2nd order Coupled Equation.

Cartesian Coordinates $\{x, y, z\}$ $T = \frac{1}{2} m (\dot{x})^2$

$$\frac{d}{dt} (m\dot{x}) + \frac{\partial V}{\partial x} = 0$$

$$\frac{\partial L}{\partial \dot{x}} = m\dot{x}$$

$$\Rightarrow m\ddot{x} + \frac{\partial V}{\partial x} = 0$$

Plane Polar Coordinate

$$x = r \cos \theta, \quad y = r \sin \theta.$$

$$T = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2) = \frac{m}{2} [\dot{r}^2 + r^2 \dot{\theta}^2]$$

$$m \frac{dr}{dt} - m r \dot{\theta}^2 \quad \boxed{\frac{\partial L}{\partial r} = m r \dot{\theta}^2}$$

$$[F_r = m\ddot{r} - m r \dot{\theta}^2]$$

$$\frac{\partial L}{\partial r} = m \dot{\theta}^2$$

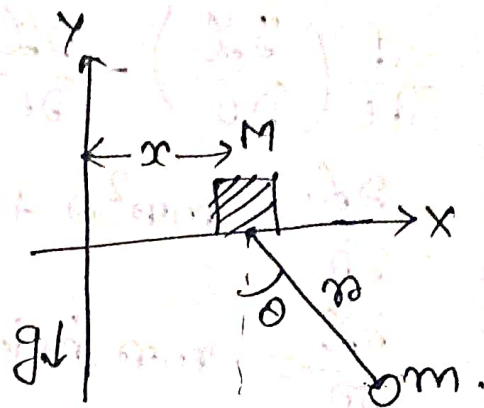
$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = m r^2 \dot{\theta} \quad \frac{\partial \mathcal{L}}{\partial \theta} = 0$$

$$\frac{d}{dt} (m r^2 \dot{\theta}) = 2 m r \dot{r} \dot{\theta} + m r^2 \ddot{\theta} = F_{\theta}.$$

$$2 m r \dot{r} \dot{\theta} + m r^2 \ddot{\theta} = F_{\theta}.$$

Gen

Generalized coordinates
are $\{x, \theta\}$.



kinetic energy of M :- $\frac{1}{2} M \dot{x}^2$
(T_M)

For the small mass m ,
displacement $\rightarrow$

$$x = r \sin \theta, \quad \dot{x} = r \cos \theta \dot{\theta}$$

$$y = r \cos \theta, \quad \dot{y} = -r \sin \theta \dot{\theta}$$

~~$\frac{1}{2} m \dot{x}^2 + \frac{1}{2} m \dot{y}^2$~~
 $x = r \sin \theta$
 $y = r \cos \theta$

$T_m = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2)$

~~$\frac{1}{2} m$~~

$\dot{x}' = \dot{x} + r \cos \theta \dot{\theta}$
 $\dot{y}' = -r \sin \theta \dot{\theta}$

$$\Rightarrow T_m = \frac{1}{2} m (\dot{x}^2 + 2r \cos \theta \dot{x} \dot{\theta} + r^2 \cos^2 \theta \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\theta}^2)$$

$$= \frac{1}{2} m (\dot{x}^2 + r^2 \dot{\theta}^2 + 2r \cos \theta \dot{x} \dot{\theta})$$

Potential energy of m (V_m) = $-mg r \cos \theta$

$\mathcal{L} = T - V$ $T = T_m + T_M$

$$\Rightarrow \mathcal{L} = \frac{1}{2} (M+m) \dot{x}^2 + \frac{1}{2} m (r^2 \dot{\theta}^2 + 2r \dot{x} \cos \theta \dot{\theta})$$

$$+ mg r \cos \theta$$

$\mathcal{L}(x, \dot{x}, \theta, \dot{\theta})$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) - \frac{\partial \mathcal{L}}{\partial x} = 0$$

$$\frac{\partial \mathcal{L}}{\partial \dot{x}} = (M+m) \dot{x} + m r \cos \theta \dot{\theta}, \quad \frac{\partial \mathcal{L}}{\partial x} = 0$$

$$\Rightarrow \frac{d}{dt} \left[(M+m) \dot{x} + mn \cos \theta \dot{\theta} \right] = 0$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) - \frac{\partial \mathcal{L}}{\partial \theta} = 0$$

$$\hookrightarrow (M+m) \dot{x} + mn \cos \theta \dot{\theta} = \text{constant} \quad (\text{Ans})$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = mn^2 \dot{\theta} + mn \dot{x} \cos \theta$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = -mn \dot{x} \sin \theta - mgn \sin \theta$$

$$\Rightarrow \frac{d}{dt} (mn^2 \dot{\theta} + mn \dot{x} \cos \theta) + mn \dot{x} \sin \theta \dot{\theta} + mgn \sin \theta = 0$$

$$\Rightarrow mn^2 \ddot{\theta} + mn \ddot{x} \cos \theta - mn \dot{x} \sin \theta \dot{\theta} + mn \dot{x} \sin \theta \dot{\theta} + mgn \sin \theta = 0$$

$$\Rightarrow mn^2 \ddot{\theta} + mn \ddot{x} \cos \theta + mgn \sin \theta = 0$$

$$\Rightarrow n \ddot{\theta} + \ddot{x} \cos \theta + g \sin \theta = 0 \quad (\text{Ans})$$

$$2) \mathcal{L} = T - V$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} = 0 \quad \mathcal{L}' = \mathcal{L} + \frac{dF(q,t)}{dt}$$

$$\mathcal{L}' = \mathcal{L} + \frac{\partial F}{\partial q_j} \dot{q}_j + \frac{\partial F}{\partial t}$$

$$\frac{\partial \mathcal{L}'}{\partial \dot{q}_j} = \frac{\partial \mathcal{L}}{\partial \dot{q}_j} + \frac{\partial F}{\partial q_j}$$

$$\frac{\partial \mathcal{L}'}{\partial q_j} = \frac{\partial \mathcal{L}}{\partial q_j} + \frac{\partial^2 F}{\partial q_k \partial q_j} \dot{q}_j + \frac{\partial^2 F}{\partial q_j \partial t} \Rightarrow \frac{d}{dt} \left(\frac{\partial \mathcal{L}'}{\partial \dot{q}} \right) - \frac{\partial \mathcal{L}'}{\partial q} = \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right) - \frac{\partial \mathcal{L}}{\partial q}$$

Gauge Degree of Lagrangian:-

LEOM is invariant under transformation $\mathcal{L}' = \mathcal{L} + \frac{dF(q,t)}{dt}$

where $F(q,t)$ is a function of (q,t) only.

Ex:- $\mathcal{L} = \frac{1}{2} q \dot{q}$, $F = -\frac{1}{4} q^2$

$$\Rightarrow \frac{dF}{dt} = -\frac{1}{2} q \dot{q} \Rightarrow \mathcal{L}' = 0 \rightarrow \text{Not a Lagrangian}$$

$$L = T - V$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) = -Q_j$$

$$-\frac{\partial V}{\partial q_j} = Q_j$$

$$\frac{dP_j}{dt} = Q_j$$

Generalized momentum $P_i = \frac{\partial L}{\partial \dot{q}_i}$

Ex:-

$$L = \frac{1}{2} m \dot{x}^2$$

$$\frac{\partial L}{\partial \dot{x}} = m \dot{x} = P_x$$

$$L = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2)$$

$$P_r = m \dot{r}, P_\theta = m r^2 \dot{\theta}$$

In a Lagrangian system -

$$\mathcal{L}(q, \dot{q}, t) \Rightarrow p_i = \frac{\partial \mathcal{L}}{\partial \dot{q}_i}$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = 0.$$

$$\text{if } q_j, \frac{\partial \mathcal{L}}{\partial q_j} = 0.$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) = 0$$

$$p_j = \frac{\partial \mathcal{L}}{\partial \dot{q}_j} = \text{constant}.$$

cyclic coordinate.

- If Lagrangian is independent of a specific generalized coordinate $q_i \Rightarrow \frac{\partial \mathcal{L}}{\partial q_i} = 0$, corresponding generalized p_i is constant of motion.

$$\frac{dp_j}{dt} = 0.$$

q_j is called cyclic coordinate.

$$\mathcal{L} = \frac{1}{2} m \dot{x}^2 \quad \frac{\partial \mathcal{L}}{\partial \dot{x}} = p_x = \text{constant}.$$

$$\mathcal{L} = \frac{1}{2} (m \dot{r}^2 + r^2 \dot{\theta}^2)$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = 0 \quad p_\theta = m r^2 \dot{\theta}$$

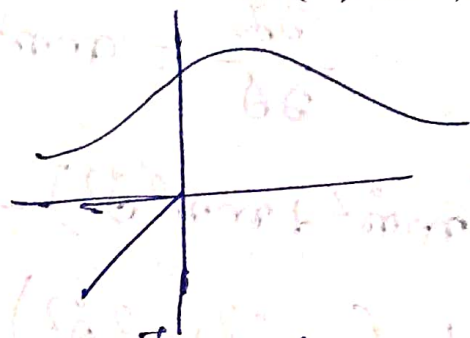
can these be

$$\frac{d\mathcal{L}}{dt} = 0 \text{ ?}$$

$$\mathcal{L}(q, \dot{q}, t).$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} = 0.$$

$$\mathcal{L}(q, \dot{q}, t) = \text{const.}$$



$$\frac{d\mathcal{L}}{dt} = \sum_i \frac{\partial \mathcal{L}}{\partial q_i} \dot{q}_i + \sum_i \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \frac{d}{dt}(\dot{q}_i) + \frac{\partial \mathcal{L}}{\partial t}.$$

$$= \sum_j \left[\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) \dot{q}_j + \frac{\partial \mathcal{L}}{\partial \dot{q}_j} \frac{d}{dt}(\dot{q}_j) \right] + \frac{\partial \mathcal{L}}{\partial t}.$$

$$\frac{d\mathcal{L}}{dt} = \frac{d}{dt} \left(\sum \dot{q}_j \frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) + \frac{\partial \mathcal{L}}{\partial t}.$$

$$\frac{d}{dt} \left[\left(\sum \dot{q}_j \frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \mathcal{L} \right] = - \frac{\partial \mathcal{L}}{\partial t}.$$

$$h = \sum \dot{q}_j \frac{\partial \mathcal{L}}{\partial \dot{q}_j} - \mathcal{L}.$$

$$h = \sum \dot{q}_j p_j - \mathcal{L}.$$

h is called energy function

$$\frac{dh}{dt} = - \frac{\partial \mathcal{L}}{\partial t}.$$

If Lagrangian $\mathcal{L}_0$ is not an explicit function of time, then energy function h is constant of motion.

$$h = \sum \dot{q}_i \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \mathcal{L}.$$

$$\mathcal{L} = \frac{1}{2} [m\dot{r}^2 + mr^2 \dot{\theta}^2] - V(r).$$

$$\frac{\partial \mathcal{L}}{\partial t} = 0, \quad \frac{\partial \mathcal{L}}{\partial \dot{r}} = m\dot{r}.$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = mr^2 \dot{\theta}$$

$$h = (m\dot{r}^2 + mr^2 \dot{\theta}^2) - \frac{1}{2} m (\dot{r}^2 + mr^2 \dot{\theta}^2) + V.$$

$$h = \frac{1}{2} (m\dot{r}^2 + mr^2 \dot{\theta}^2) + V.$$

$$= T + V = E \rightarrow \text{Total energy}$$

$$\mathcal{L} = T(q, \dot{q}, t) - V(q).$$

In Euclidean coordinates -

$$T = \sum_i \frac{1}{2} m_i v_i^2.$$

$$\vec{v}_i = \frac{d\vec{r}_i}{dt}$$

$$\vec{v}_i = \sum_k \frac{\partial \vec{r}_i}{\partial q_k} \dot{q}_k + \frac{\partial \vec{r}_i}{\partial t}$$

$$\begin{aligned} \sum_i \frac{1}{2} m_i v_i^2 &= \sum_i \frac{1}{2} m_i \left(\sum_j \frac{\partial \vec{r}_i}{\partial q_j} \dot{q}_j + \frac{\partial \vec{r}_i}{\partial t} \right)^2 \\ &= \sum_i \frac{1}{2} m_i \sum_{j,k} \left(\frac{\partial \vec{r}_i}{\partial q_j} \dot{q}_j \right) \left(\frac{\partial \vec{r}_i}{\partial q_k} \dot{q}_k \right) \\ &\quad + m_i \frac{\partial \vec{r}_i}{\partial t} \sum_j \frac{\partial \vec{r}_i}{\partial q_j} \dot{q}_j + \frac{1}{2} m_i \frac{\partial \vec{r}_i}{\partial t} \cdot \frac{\partial \vec{r}_i}{\partial t} \end{aligned}$$

$$= \frac{1}{2} \sum_{j,k} M_{jk} \dot{q}_k \dot{q}_j + \sum_j M_j \dot{q}_j + M_0$$

$$M_{ij} = \sum_i m_i \frac{\partial \vec{r}_i}{\partial q_j} \cdot \frac{\partial \vec{r}_i}{\partial q_k}.$$

$$M_j = \sum_i m_i \frac{\partial m_i}{\partial q_j} \frac{\partial m_i}{\partial t}$$

$$M_0 = \frac{1}{2} \sum m_i \left(\frac{\partial m_i}{\partial t} \right)^2$$

$$T = T_0 + T_1 + T_2$$

$T_0 \Rightarrow$ function of q, t only.

$T_1 \Rightarrow$ linear homogeneous function in $\dot{q}$.

Coefficient can be function of q and t .

$T_2 \Rightarrow$ 2nd order homogeneous function in $\dot{q}_i$.

$$L = T - V(q_i)$$

$$L(q, \dot{q}, t) = L_0(q, t) + L_1(q, \dot{q}, t) + L_2(q, \dot{q}, t).$$

$$L = L_0 + L_1 + L_2.$$

L_0 : is function of q, t .

L_1 : is function of q, t and linear homogeneous in $\dot{q}_i$.

L_2 : is ^{also} function of q, t and second order homogeneous in $\dot{q}_i$.

If $f(x_i)$ is homogeneous function of order n .

Then we have -

$$\sum_i x_i \frac{\partial f}{\partial x_i} = n f$$

$$h = \sum_i \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} - L.$$

from here.

$$L = L_0 + L_1 + L_2$$

$$h = \sum_i \dot{q}_i \frac{\partial}{\partial \dot{q}_i} (L_0 + L_1 + L_2) - (L_0 + L_1 + L_2).$$

$$h = L_1 + 2L_2 - L_0 - L_1 - L_2.$$

$$h = L_2 - L_0.$$

When h becomes ^{total} energy? his a function of energy (energy function).
_{condition}

$$\Rightarrow \text{When } \boxed{L_2 = T, L_0 = -V.}$$

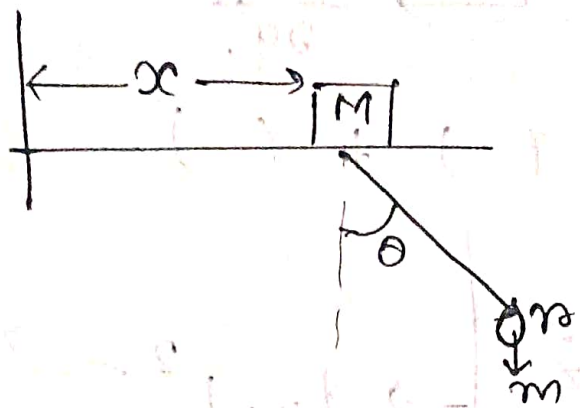
$$\text{and } T = \vec{p}_0^0 + \vec{p}_1^0 + T_2 \Rightarrow T = T_2.$$

$$L = \frac{1}{2} \sum () \dot{q}_i \dot{q}_j \quad L_0(q, t) = -V.$$

$$T = \frac{1}{2} \sum_i m_i v_i^2$$

Example :-

1)



Lagrangian (done previously) —

$$\mathcal{L}(x, \dot{x}, \theta, \dot{\theta}) = \frac{1}{2} (M+m) \dot{x}^2 + \frac{m}{2} (r^2 \dot{\theta}^2 + 2r \dot{\theta} \dot{x} \cos \theta) + mgr \cos \theta$$

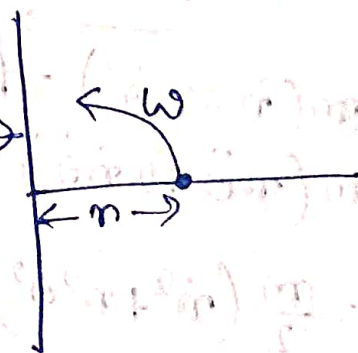
$$T = \frac{1}{2} (M+m) \dot{x}^2 + \frac{m}{2} (r^2 \dot{\theta}^2 + 2r \dot{\theta} \dot{x} \cos \theta)$$

$$T = T_2 - \mathcal{L}_2$$

$$\mathcal{L}_0 = -V = mgr \cos \theta$$

$$h = T + V$$

2) Example from Goldstein.



$$T = \frac{1}{2} m \dot{r}^2 + \frac{1}{2} m r^2 \omega^2 = T_0 + T_2$$

$$V = 0$$

ω is constant.

$$h = \mathcal{L}_2 - \mathcal{L}_0 = \frac{1}{2} m \dot{r}^2 - \frac{1}{2} m r^2 \omega^2$$

$$h = \sum \dot{q}_i \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \mathcal{L}$$

$$E = T + V = \frac{1}{2} m \dot{r}^2 + \frac{1}{2} m r^2 \omega^2$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right) - \frac{\partial \mathcal{L}}{\partial q} = 0$$

$$\rightarrow \frac{\partial V_{eff}}{\partial q}$$

$$h = \sum \dot{q}_i \dot{q}_j + () + ()$$

$$L_x = y \dot{z} - z \dot{y}$$

$$L_y = z \dot{x} - x \dot{z}$$

$$L_z = x \dot{y} - y \dot{x}$$

$$L_z, L^2$$

they commute

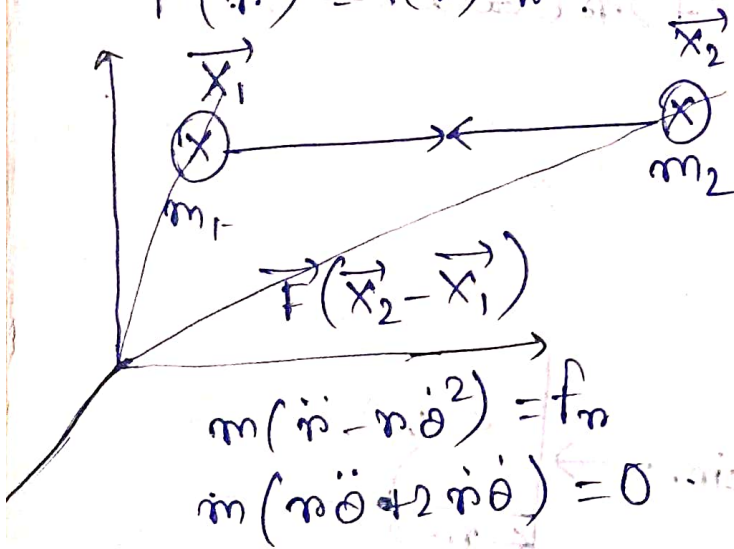
conserved and function of quadratic w.r.t. in velocity.

27/08/2025

Central Force Problem

$$\vec{F}(\vec{r}) = f(r) \hat{r}$$

$$\vec{F}(\vec{r}) = -\nabla V = f(r) \hat{r} = V(r)$$



$$\mathcal{L} = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\theta}^2) - V(r)$$

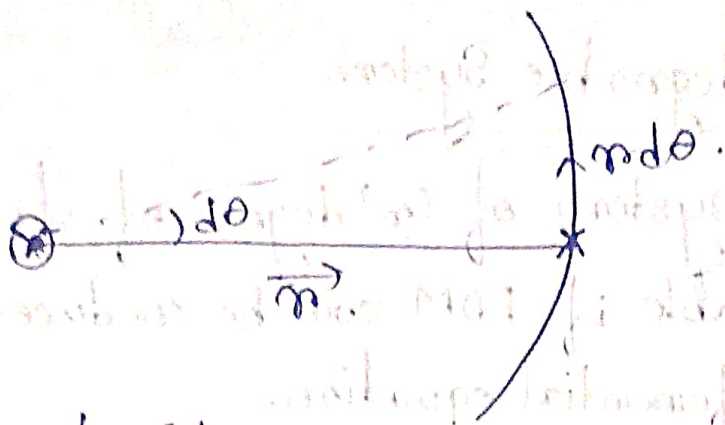
$$\frac{\partial \mathcal{L}}{\partial \theta} = 0 \quad \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = L_z = m r^2 \dot{\theta} = \text{constant}$$

$$\dot{\theta} = \frac{L_z}{m r^2}$$

$$\frac{d}{dt} (m r^2 \dot{\theta}) = 0$$

$$\frac{d}{dt} m \frac{d}{dt} \left(\frac{1}{2} r^2 \dot{\theta} \right) = 0$$

$$\text{Angular velocity} = \frac{d\theta}{dt}$$



$$dA = \frac{1}{2} r^2 d\theta$$

$$\Rightarrow \frac{dA}{dt} = \frac{1}{2} r^2 \dot{\theta}$$

$$\Rightarrow \frac{dA}{dt} = \text{constant}$$

$$V(r) = -\frac{GMm}{r} \Rightarrow \text{Kepler's Problem}$$

$$\frac{\partial \mathcal{L}}{\partial t} = 0, \quad h = L_2 - L_0$$

$$L_2 = T, \quad L_0 = -V$$

$$h = T + V = E$$

$$E = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\theta}^2) + V(r)$$

$$\frac{2}{m} (E - V) = \dot{r}^2 + r^2 \dot{\theta}^2$$

$$= \dot{r}^2 + \frac{L_z^2}{m^2 r^2}$$

$$\dot{r}^2 = \frac{2}{m} (E - V) - \frac{L_z^2}{m^2 r^2}$$

Integrable System

If a Lagrangian system of (n) degree of freedom is said to be integrable if EOM can be reduced to (n) first order differential equation.

Corollary:- If Lagrangian system has (n) independent constant of motion then it is integrable.

$$\dot{r} = \pm \left[\frac{2}{m} (E - V) - \frac{L_z^2}{m^2 r^2} \right]^{\frac{1}{2}}$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{r}} \right) - \frac{\partial \mathcal{L}}{\partial r} = 0.$$

$$\frac{\partial \mathcal{L}}{\partial \dot{r}} = m \dot{r} \quad \frac{\partial \mathcal{L}}{\partial r} = m r \dot{\theta}^2 - \frac{\partial V}{\partial r}$$

$$m \ddot{r} - m r \dot{\theta}^2 + \frac{\partial V}{\partial r} = 0.$$

$$\Rightarrow m \ddot{r} - \frac{L_z^2}{m r^3} + \frac{\partial V}{\partial r} = 0$$

$$\Rightarrow m \ddot{r} = \frac{L_z^2}{m r^3} - \frac{\partial V}{\partial r} = - \frac{\partial}{\partial r} \left(\frac{L_z^2}{2 m r^2} + V \right).$$

$$m \ddot{r} = F_r = - \nabla V_{\text{eff}}.$$

$$V_{\text{eff}} = \frac{L_z^2}{2 m r^2} + V.$$

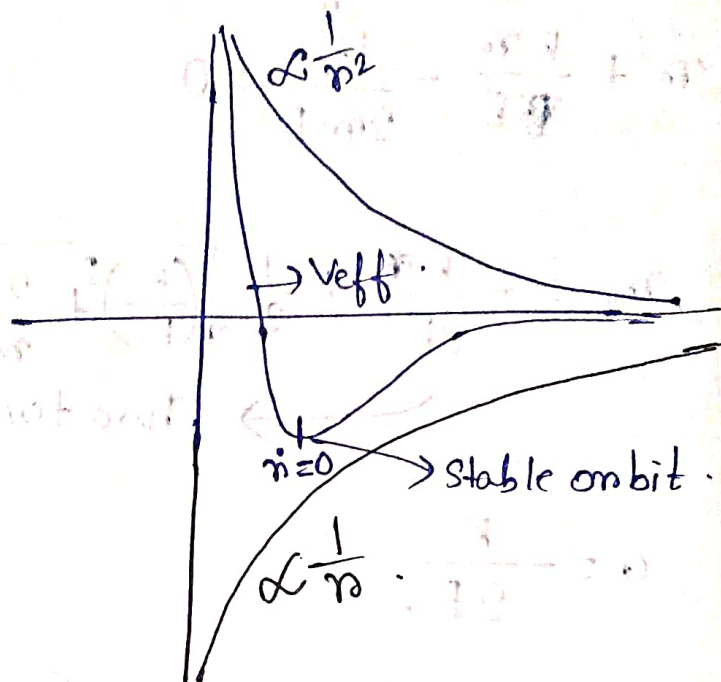
$$\dot{r} = \pm \left[\frac{2}{m} \left(E - V - \frac{L_z^2}{2 m r^2} \right) \right]^{\frac{1}{2}}$$
$$= \pm \left[\frac{2}{m} (E - V_{\text{eff}}) \right]^{\frac{1}{2}}$$

Kepler Problem

$$V_{\text{eff}} = \frac{L_z^2}{2mr^2} - \frac{k}{r}$$

$$V_{\text{kep}} = -\frac{k}{r}$$

$$k = G M m$$



29/08/2025

Central Force Problem

$$\mathcal{L} = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\theta}^2) - V(r)$$

$$m r^2 \dot{\theta} = L_z \Rightarrow \dot{\theta} = \frac{L_z}{m r^2}$$

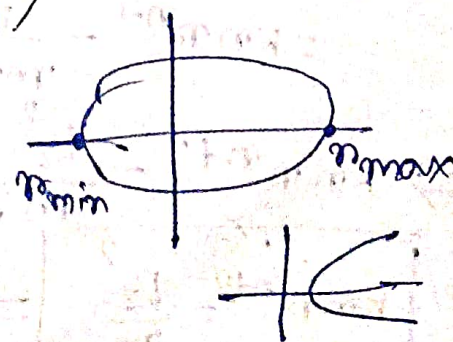
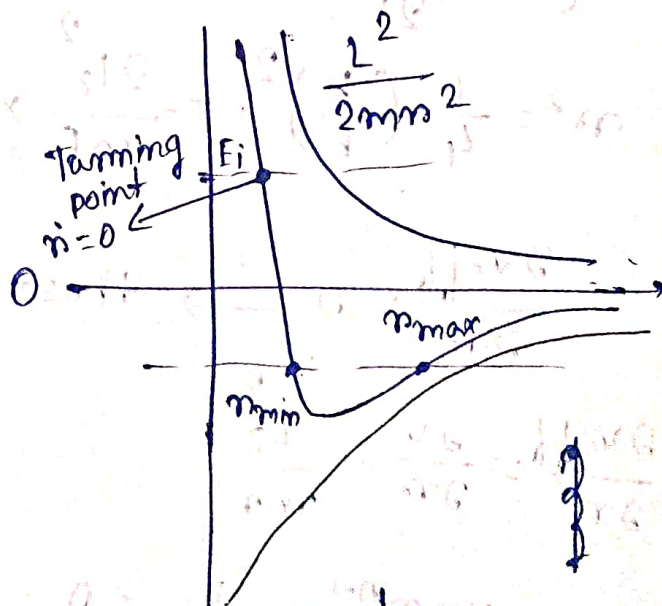
$$\dot{r} = \left[\frac{2}{m} \left(E - V - \frac{L_z^2}{2m r^2} \right) \right]^{\frac{1}{2}}$$

$$V_{\text{eff}} = V + \frac{1}{2} \frac{L_z^2}{m r^2}$$

$$E = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2) + V$$

Kepler problem: $V(r) = -\frac{k}{r}$

Semi major axis $= (r_{\text{min}} + r_{\text{max}}) \times \frac{1}{2} = a$



$$1 + \frac{k}{r} - \frac{L_z^2}{2mr^2} = 0$$

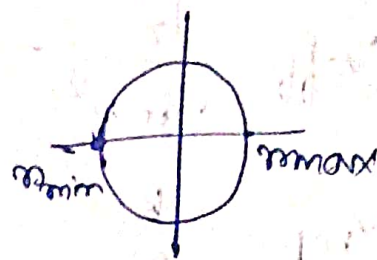
$$r^2 + \frac{k r}{E} - \frac{L_z^2}{2mE} = 0$$

$$r_{\pm} = -\frac{k}{2E} \pm \frac{1}{2} \sqrt{\left(\frac{k}{E}\right)^2 + \frac{2L_z^2}{mE}}$$

Two turning point.

$$a = -\frac{k}{2E}$$

Circular Orbit



$$r_{\max} = r_{\min}$$

For circular orbit (r_c):-

$$r_+ = r_- \Rightarrow \frac{k^2}{E^2} = -\frac{2L_z^2}{mE}$$

We have taken $E < 0$.

$$\frac{L_z^2}{m k}$$

$$r_c = -\frac{k}{2E}$$

$$r^2 = \frac{1}{4} \left(\frac{k}{E}\right)^2 = -\frac{2L_z^2}{mE} \times \frac{1}{4} = -\frac{1}{2} \frac{L_z^2}{mE}$$

check whole derivation once.

$\frac{\partial V_{eff}}{\partial r} = 0 \rightarrow$ these are the optimum points.

$$\begin{aligned} \frac{\partial V_{eff}}{\partial r} &= \frac{\partial V}{\partial r} - \frac{L_z^2}{mr^3} \\ &= kmr^{n-1} - \frac{L_z^2}{mr^3} = 0 \end{aligned}$$

$$\Rightarrow kmr^{n+2} = \frac{L_z^2}{m}$$

when $k = -\frac{GMm}{r}$ we go back to the Kepler problem.

$$\boxed{kmr^{n+2} = \frac{L_z^2}{m}} \Rightarrow \text{condition for circular orbit.}$$

$$V \propto km^n$$

Any function can be expressed as the power of r^n .

For stable circular orbit

$$\frac{\partial^2 V}{\partial r^2} > 0$$

$$\Rightarrow km(n-1)r^{n-2} + \frac{3L_z^2}{mr^4} > 0$$

$$\Rightarrow km(n-1)r^{n+2} + \frac{3L_z^2}{m} > 0$$

$$\frac{L_z^2}{m} = kmr^{n+2}$$

$$\Rightarrow km(n-1)r^{n+2} + 3kmr^{n+2} > 0$$

$$\Rightarrow (n-1+3)\frac{L_z^2}{m} > 0$$

$$\Rightarrow (n+2)\frac{L_z^2}{m} > 0$$

Orbit equation $r(\theta)$

$\theta(t), r(t)$

$$\frac{dr}{dt} / \frac{d\theta}{dt} = \frac{dr}{d\theta}$$

$$\frac{d\theta}{dt} = \frac{L_z}{mr^2} \Rightarrow dt = mr^2 \frac{d\theta}{L_z}$$

$$m\ddot{r} = -\frac{\partial}{\partial r} \left[V + \frac{1}{2} \frac{L_z^2}{mr^2} \right] = -\frac{d}{dr} \left[V + \frac{1}{2} \frac{L_z^2}{mr^2} \right]$$

$$\frac{d}{dt} \left(m \frac{dr}{dt} \right) \frac{d\theta}{dt} = -\frac{d}{dr} \left[V + \frac{1}{2} \frac{L_z^2}{mr^2} \right] \frac{d\theta}{dt} \frac{dt}{d\theta}$$

$$\Rightarrow \frac{d}{d\theta} \left(\frac{1}{r^2} \frac{dr}{d\theta} \right) = \frac{d}{dr} \left[V + \frac{1}{2} \frac{L_z^2}{mr^2} \right] \frac{1}{L_z}$$

$$U = \frac{1}{r}, \quad \frac{dU}{d\theta} = -\frac{1}{r^2} \frac{dr}{d\theta} = \frac{dU}{dr} \frac{dr}{d\theta}$$

By calculating explicitly we will get -

$$\frac{d}{d\theta} \left(\frac{1}{r} \frac{dU}{d\theta} \right) = \frac{mk}{L_z^2} - U$$

$$\frac{1}{u^2}$$

Homework :-

Show that

$$m\ddot{r} = \frac{d}{dr} \left[-\frac{k}{r} + \frac{L_z^2}{2mr^2} \right]$$

Change $\frac{dr}{dt}$ to $\frac{dr}{d\theta}$.
and $u \Rightarrow \frac{1}{r}$

Finally we have,

$$\frac{d}{d\theta} \left(\frac{du}{d\theta} \right) + u = \frac{mk}{L_z^2}$$

$$\Rightarrow \frac{d^2 u}{d\theta^2} + u = \frac{mk}{L_z^2}$$

$$\Rightarrow u = A \cos(\theta - \theta_0) + \frac{mk}{L_z^2} \quad \left(\text{Harmonic Oscillation type equation} \right)$$

$$\frac{1}{r} = \frac{mk}{L_z^2} \left[1 + \sqrt{\frac{1 + 2EL_z^2}{mk^2}} \cos(\theta - \theta_0) \right]$$

$$\frac{1}{r} = C [1 + e \cos(\theta - \theta_0)] \rightarrow \text{equation of ellipse}$$

$$e = \sqrt{\frac{1 + 2EL_z^2}{mk^2}} = \text{eccentricity of ellipse}$$

Homework :- (Next Friday).

Show that bound orbit in keplers problem are ellipses. Find the expression for e in terms of l_z, E .